

Codebase Cost Estimation Report

Repository: cost-estimate **Analysis Date:** 2026-03-06 **Methodology:** Simplified
Parametric Model based on COCOMO II

Codebase Profile

Metric	Value
Languages	Markdown (1,278), Shell (29)
Total Lines of Code	1,307
Effective KLOC	0.03 KLOC (source code only)
Configuration LOC	1,278 (Markdown – reported for context, not used in COCOMO)
Total Files	9
Git Commits	1
Contributors	1 (Margus Lamp)
Repository Age	New (same day)
Active Development	2026-03-06 □ 2026-03-06
Tech Stack	Shell/Bash, Markdown documentation, Claude Code skill

Note: This codebase is under 500 effective LOC. The parametric model is calibrated for projects above 2 KLOC. The estimates below have low confidence and should be treated as rough order-of-magnitude only.

Complexity Assessment

Factor	Score	Justification
External Integrations	1/5	No HTTP clients or API calls found
Data Layer	1/5	No database or ORM patterns found
Auth & Authorization	1/5	No auth patterns found
Testing Maturity	1/5	No test files found
Infrastructure/DevOps	1/5	No Docker, CI/CD, or IaC files found
Error Handling	1/5	No observability or error handling patterns
Documentation	2/5	README.md present with usage instructions
Security Posture	1/5	No security patterns found
Average	1.1/5	Simple

Intellectual Effort Artifacts

Non-code files representing significant domain expertise, prompt engineering, or encoded methodology.

Classification	Files	Physical Lines	Equiv. Effort LOC	Key Files
Domain Expertise (T3, 1.5x)	1	1,397	2,096	SKILL.md
Boilerplate (T1, 0.1x)	2	245	24	(not listed)
Total	3	1,642	2,120	

Intellectual effort artifacts add **2,096** equivalent LOC to the project, representing **98.6%** of total estimated effort.

Effort Estimation

Parameter	Value
Effective KLOC	0.03 (Shell source code)
Project Type	Simple (avg complexity 1.1/5)
Exponent E	1.06
Effort Multipliers	CPLX=0.73, DATA=0.90, RELY=0.82, PLEX=0.87, DOCU=0.91
EAF (product of EMs)	0.43
Source Code Effort	0.0 person-months (4 hours)
Intellectual Effort Artifacts	2.1 person-months (323 hours)
Combined Base Effort	2.2 person-months (330 hours)
Estimated Function Points	53 FP (1 Source Code + 52 IE)
Schedule (Tdev)	4.6 months (ideal)

Effort Range

	Low (0.6x)	Mid (1.0x)	High (1.6x)
Person-Months	1.3	2.2	3.4
Person-Hours	200	330	520

Cost Estimation

By Team Configuration

Team Profile	Low	Mid	High	Calendar Time (Mid)
Solo Senior (\$175/hr)	\$34K	\$57K	\$92K	4.6 months
Lean Startup (\$150/hr, 1.15x OH)	\$34K	\$56K	\$90K	4.6 months
Growth Co (\$165/hr, 1.35x OH)	\$44K	\$73K	\$117K	4.6 months
Enterprise (\$185/hr, 1.65x OH)	\$60K	\$100K	\$160K	4.6 months

Effort Allocation (Mid Estimate)

Activity	%	Hours	Cost (Growth Co)
Development & Coding	55%	180	\$40K
Testing & QA	20%	70	\$16K
Project Management	12%	40	\$9K
DevOps & Infrastructure	8%	30	\$7K
Documentation	5%	20	\$4K

Headline Valuation

Tier	Estimate	Basis
Conservative	\$34K	Solo senior, low complexity (0.6x)
Realistic	\$73K	Growth company, mid estimate
Premium	\$160K	Enterprise team, high estimate (1.6x)

If someone asked “what would it cost to build this from scratch?” The realistic answer is **\$44K - \$117K** with a team of 6.5 engineers over 4.6 months. This includes **323 hours** of intellectual effort in non-code artifacts (prompts, domain configs, methodology documents).

AI Speed Comparison

Metric	Human Team	AI-Assisted
Build Time	330 hours	3 hours
Speed Multiple	1x	110x faster
Cost (at Growth Co rates)	\$73K	\$700
Cost Savings	—	99%

AI-assisted development completed in **AI 2h Claude 4.6, human 1h** what would take a human team **330 hours** – a **110x speedup**.

Methodology & Assumptions

- **Model:** Simplified parametric model based on COCOMO II (Boehm et al., 2000). Uses the COCOMO II base equation ($A=2.94$, exponent E from Scale Factors) with 5 of 17 Effort Multipliers mapped from automated codebase analysis; remaining EMs default to Nominal (1.0). This is an automated approximation – full COCOMO II requires expert calibration of all 22 drivers.
- **Working Hours:** 152 hours/month (19 productive days x 8 hours, accounting for holidays and administrative time)
- **LOC Measurement:** cloc (code lines only, excluding comments and blanks)
- **Rates:** Fully-loaded US consulting/agency rates (2025-2026), not salary equivalents

- **Overhead:** Accounts for hiring/onboarding, organizational coordination, compliance processes, and institutional overhead beyond direct development (see Team Profiles note)
- **Exclusions:** Generated code, lock files, vendored dependencies, build artifacts, config/markup files (from KLOC)
- **Intellectual Effort:** Non-code artifacts (prompts, domain configs, rubrics) estimated using a 4-tier classification system (Boilerplate 0.1x, Structured Knowledge 0.5x, Domain Expertise 1.5x, Novel Methodology 3.0x) based on automated signal detection (conditional language density, constraint patterns, instructional density). Constants derived from Jones, *Applied Software Measurement* (3rd ed., 2008) and IFPUG SNAP Assessment Practices Manual v2.4 (2017). Intellectual effort estimation is a supplementary methodology, not part of standard COCOMO II.
- **Conservative bias:** Estimates lean toward underestimation; actual costs often exceed parametric predictions for novel or poorly-defined projects
- **Schedule:** COCOMO II Tdev formula: $Tdev = 3.67 * PM^F$ where $F = 0.28 + 0.2*(E-0.91)$; Calendar Time is the greater of naive parallelization (hours / headcount / 152) and Tdev, since Tdev represents the minimum feasible schedule regardless of staffing
- **Small project caveat:** Model is calibrated for 2+ KLOC projects; sub-500 LOC estimates are low-confidence

Sources: Boehm, B. et al. (2000). *Software Cost Estimation with COCOMO II*. Prentice Hall. Jones, C. *Applied Software Measurement*, 3rd ed. McGraw-Hill. ZipRecruiter, FullStack Labs, Rise 2026 Contractor Rates.

Generated by [cost-estimate](#) – a Claude Code skill using a simplified parametric model based on COCOMO II